

Problem Set – Introduction to Functions.

1. Allow the user to repeatedly enter a quantity and price. Prompt the user on whether they want to do the program (Yes or No). Use a function to compute the total (quantity times price). The function should be passed the quantity and price and then return the total. In the function, provide a 10% discount if the total is over \$10,0000.00. Display quantity, price and total. Sum and display the extended price.

Input	Process	Output
Quantity (float)	Compute extended price (quantity, unit price) 1. Extended price = quantity * unit price 2. If extended price > 100000.00 then - Discount = extended price * 0.10 - Else - Discount = 0.0 3. Return extended price – discount	- Quantity - Unit price - Extended price
Unit Price (float)	Main 1. Total extended price = 0.0 2. Prompt “Do you want to compute extended price? (yes/no)” – choice 3. While choice == ‘yes’ - Read quantity, unit price - Call compute extended price – extended price - Display quantity, unit price, extended price - Total extended price += extended price - Prompt “Continue? (yes/no)” – choice - End loop	Total extended price
	Display total extended price	

2. Enter players last name, number of hits and at bats at the keyboard. Prompt the user on whether they want to do the program (Yes or No). Use a function to compute batting average. Pass the hits and at bats to the function. The function should return batting average. Display last name and batting average. Give a count of the number of players entered.

Input	Process	Output
	Compute batting average (hits, at bats) Average = hits/at bats if at bats>0 Else 0.0 Return average	
• Last name • Hits • At bats	Main Player count = 0 Choice = get yes no ("Please enter player data? (yes/no):") While choice == 'y': • Read last name • Read hits (int), at bats (int) • Average = compute batting average (hits, at bats) • Print last name and average • Player count += 1 • Choice = get yes no ("Continue? (yes/no):") End loop	• Last name • Batting average
	Display number of players entered	• Number of players entered

3. Enter the destination city, miles travelled and gallons used for a trip. Prompt the user on whether they want to do the program (Yes or No). Use a function to compute miles per gallon and cost of gas. Pass miles travelled and gallons used to the function. The function should return miles per gallon and compute gas cost to be gallons times 3.00. Count the number of entries made (number of trips) Display destination city, miles, mpg and gas cost. At end display the number of entries made, total miles travelled for all trips and total gas cost of all trips.

Input	Process	Output
	Function Compute trip statistic (miles, gallons) 1. If gallons > 0: mpg = miles / gallons Else mpg = 0.0 2. Cost = gallons * 3.00 3. Return (mpg, cost)	
• Destination city (string) • Miles travelled (float) • Gallons used (float)	Main 1. Initialize: • Total miles = 0.0 • Total cost = 0.0 • Trip count = 0 2. Prompt "Please enter trip data? (yes/no)" – choice 3. While choice == 'y': • Call compute trip statistic (miles, gallons) – mpg, cost • Display destination city, miles, mpg, gas cost • Trip count += 1 • Total miles += miles • Total cost += cost End of the loop	• Destination city • Miles • Mpg • Gas cost
	Display number of entries, total miles travelled, total gas cost	• Number of entries • Total miles travelled • Total gas cost

- Allow the employee to enter last name, job code and hours worked. Prompt the user on whether they want to do the program (Yes or No). Use a function to determine the pay rate. Pass to this function the job code and it should return rate of pay and gross pay. Use Job code L is \$25/hr, A is \$30/hr and J is \$50/hr for respective pay rates. Compute gross pay. Give time and a half for overtime. Display last name, hours, pay rate and gross pay. Sum and display total of all gross pay.

Input	Process	Output
	Compute pay (job code, hours) - Get the pay rate by code (L – 25, A- 30, J- 50; else 0) - If hours > 40: -regular = 40 * rate - overtime = (hours -40) * rate * 1.5 - Gross = regular + overtime - Else: - Gross = hours* rate - Return rate, gross	
- Last name - Job code - Hours worked	Main - Total gross = 0.0 - Prompt “Do you want to compute pay? (yes/no)” – choice - While choice == ‘yes’: - Read last name - Read job code - Read hours (float) - Call computes pay (job code, hours) – rate, gross - Print last name, hours, rate, gross - Total gross += gross End of the loop	- Last name - Hours - Pay rate - Gross pay
	Display the value about total gross pay	Total gross pay

- Allow the user to enter student last name, credit hours and district code. Prompt the user on whether they want to do the program (Yes or No). Use a function to compute tuition owed. Charge In district (code of I) \$250 per credit hour. Out of district (code of O) is \$550 per credit hour. The function should receive credit hours and district code and return tuition owed. Display student name and tuition owed. Sum and display total of all tuition owed.

Input	Process	Output
	Function • Compute tuition (credits, code) 1. If code is "I": rate = 250 2. Else if code is "O": rate = 550 3. Else: rate = 0 4. Tuition = credits * rate 5. Return tuition	
• Last name (string) • Credit hours (integer) • District code	Main 1. Total tuition = 0.0 2. Prompt "Please enter student data? (yes/no)" – choice 3. While choice == 'y': • Read last name • Read credits • Read district code • Tuition = compute tuition (credits, district code) • Display last name, tuition • Total tuition += tuition owned End of the loop	• Last name • Tuition owned
	Display total of all tuition owned	• Total of all tuition owned